DS3040 Introduction to Deep Learning Lab

Assignment-4, Date: Feb 3, 2025

Timing: 10:00 to 12:45 AM | Max Marks: 20

Instructions

1. Submit one name.ipynb file containing all answers. The name should be [student_name] _assignment[number].ipynb
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Write justifications for your choices where necessary.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. Do not use AI or any websites except library documentations.

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1. Implement a feedforward neural network (FNN) for a classification task with two hidden layers from scratch, which includes (10)
   - a Forward propagation from scratch.
   - b Function for calculating the cross-entropy loss.
   - c Batch gradient descent with backpropagation from scratch. (Note: Use sigmoid as the activation function and hidden units of your choice in hidden layers.)
2. Load the given dataset and perform the following task. (5)
   - a Tune the hyperparameters epoch, hidden unit, and learning rate.
   - b Using the best hyperparameters, train your model and report the classification accuracy/loss for each epoch.
3. Implement the same network (with the same hyperparameters and activation function) using TensorFlow/PyTorch and compare it with your own model. Discuss the results and analyse them. (5)

Additional Question

4. Generalize the forward propagation and back propagation algorithms for any number of hidden layers and any number of hidden units.